FROM GEOMETRIC SERIES TO POWER SERIES
(from number series to function series)

Recall $\sum_{n=0}^{\infty} a q^n$ is called a GEOMETRIC SERIES. ($a, q \in \mathbb{R}$)

If $|q| < 1$ it is convergent and the sum can be calculated from the formula

$$a q^0 + a q^1 + a q^2 + \dots + a q^n = a \frac{1-q^n}{1-q} = S_n$$

The sum $\sum a q^n$ is equal to the $\lim_{n \to \infty} a \frac{1-q^n}{1-q} = a \frac{1}{1-q}$.

If we let x to change over some interval $x \in (a,b)$, then we obtain a function series of the very special form

$$\sum_{n=0}^{\infty} a_n x^n$$

(we let also a_n to change with $n \in \mathbb{N}$).

In general series of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^n$ is called a POWER SERIES with coefficiens $a_0, a_1, a_2, \dots$, x_0 called the center of the series and x - variable changing within the interval (a,b). Is this series convergent?

WHY POWER SERIES ARE IMPORTANT?

Recall Taylor formula, which provides local approximation to a function of class $\mathcal{C}^n([a,b])$ ie a function which has all derivatives up to the nth, and the nth derivative is continuous,

TAYLOR FORMULA

$$f(x) = \sum_{k=0}^{n-1} \underbrace{\frac{f^{(k)}(x_0)}{k!}}_{\substack{\text{number coefficients}\\ \text{depending on the}\\ \text{function } f}} (x-x_0)^k + \underbrace{\frac{f^{(n)}(\xi)}{n!}(x-x_0)^n}_{\substack{\text{reminder}\\ (\xi \text{ is a number}\\ \text{from the}\\ \text{interval}\\ \text{centered at } x_0)}}$$

If f is of class $\mathcal{C}^\infty$ we can write expression

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k \qquad (*)$$

If this series is convergent at each x_0 from the domain of f and it converges to the function $f(x)$ we say that expression $(*)$ is TAYLOR SERIES.

Notice that Taylor series is a special type of the POWER SERIES $\sum a_n (x-x_0)^n$, with $a_n = \frac{f^{(n)}(x_0)}{n!}$.

The formula $(*)$ is valid only locally around x_0 as a good aproximation of $f(x)$.

Expanding function to a power series is a standard method for solving ODES with variable coefficients.
It gives solutions in the form of a power series.
Power series representation can be used successfuly for computing values, drawing curves, exploring properties of the solutions.
Disadvantage is that this type of representation is valid only for differentiable functions and is valid only locally.
Later we shall learn about another series representation of a function, which can be used even for discontinuous functions and is valid not only locally, but it is restricted only for periodic functions.
It is called a Fourier series.

POINTWISE CONVERGENCE AND UNIFORM CONVERGENCE for function sequences

weaker type of convergence — stronger type of convergence

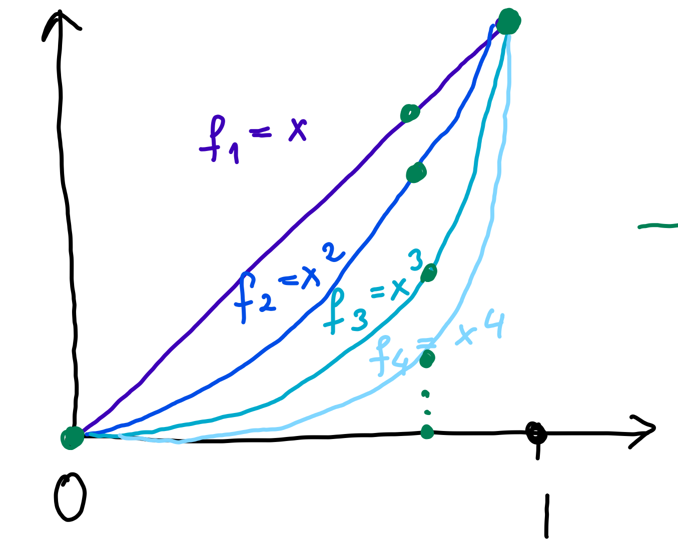
def. The sequence of functions $(f_n)_{n\in\mathbb{N}}$ is POINTWISE CONVERGENT to f

$f_n \to f$ iff $\forall \varepsilon > 0 \ \forall x \in D_{f_n} \ \exists N = N(\varepsilon, x) \ \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$

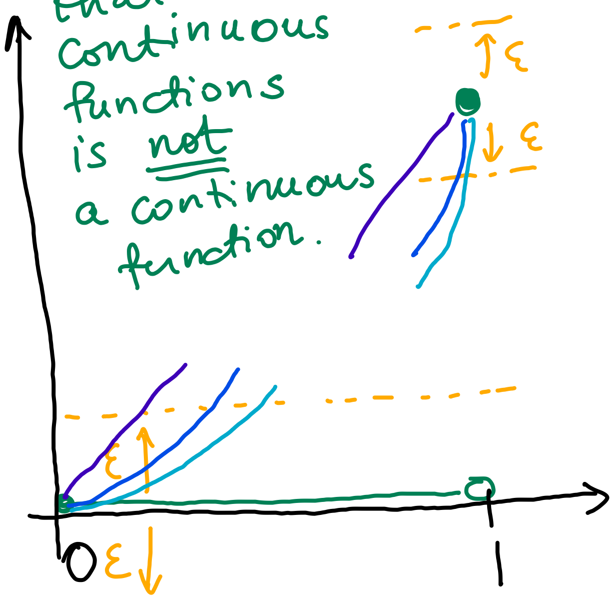
def. The sequence of functions $(f_n)_{n\in\mathbb{N}}$ is UNIFORMLY CONVERGENT to f

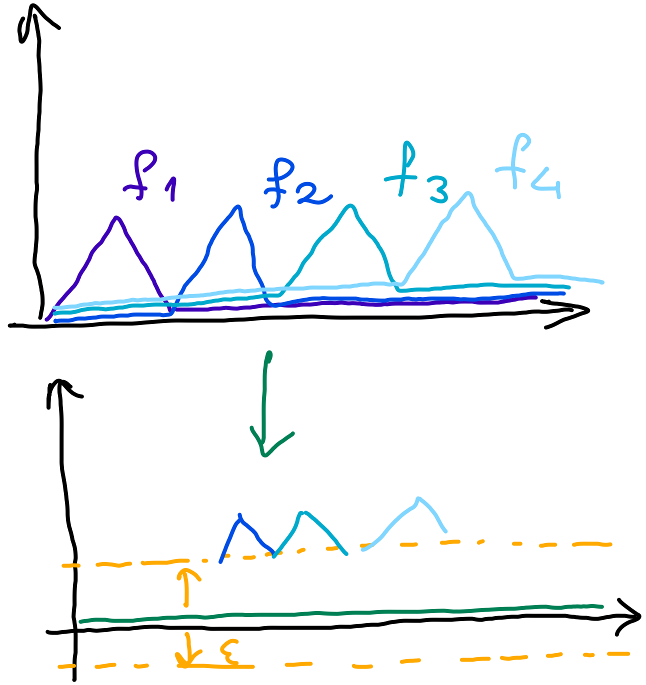
$f_n \rightrightarrows f$ iff $\forall \varepsilon > 0 \ \exists N = N(\varepsilon) \ \forall x \in D_{f_n} \ \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$

EXAMPLES



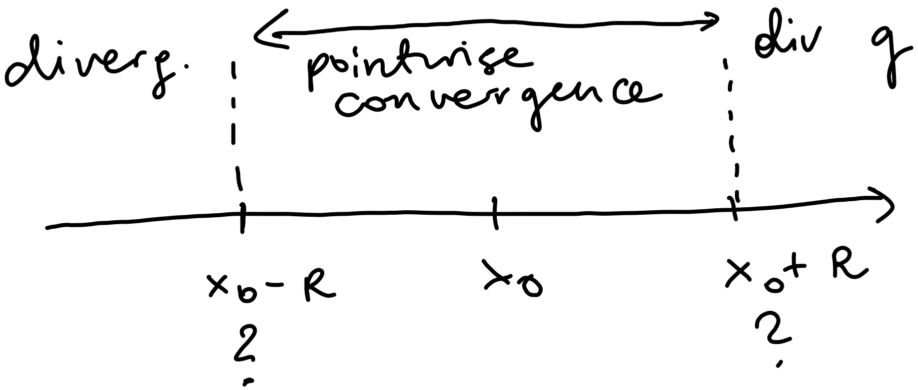
Notice in this example that the limit of continuous functions is not a continuous function.





Both examples are of pointwise convergent sequences, but not uniformly convergent.

CONVERGENCE OF A POWER SERIES



If $\sum_{n=0}^{\infty} a_n (x-x_0)^n$ is convergent at the interval (x_0-R, x_0+R) and is divergent for $x < x_0-R$ or $x > x_0+R$.

RADIUS OF CONVERGENCE $= \sup\{|x| : \sum a_n (x-x_0)^n$ is pointwise convergent at $x\}$

<u>Theorem</u>.

$$R = \begin{cases} 0, & \text{if } \lim\left|\frac{a_{n+1}}{a_n}\right| = +\infty \\ \dfrac{1}{\lim\limits_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|}, & \text{if } 0<\lim\left|\frac{a_{n+1}}{a_n}\right| < +\infty \\ +\infty, & \text{if } \lim\left|\frac{a_{n+1}}{a_n}\right| = 0 \end{cases}$$

$$R = \begin{cases} 0, & \text{if } \lim\sqrt[n]{|a_n|} = +\infty \\ \dfrac{1}{\lim\sqrt[n]{|a_n|}}, & \text{if } 0<\lim\sqrt[n]{|a_n|} < +\infty \\ +\infty, & \text{if } \lim\sqrt[n]{|a_n|} = 0 \end{cases}$$

Rem. Power series is absolutely and uniformly convergent at each closed interval $[x_0-r, x_0+r] \subset (x_0-R, x_0+R)$.

Examples from previous [illegible] show that the series may not be uniformly convergent at the entire convergence interval.

Examples

1.) $\sum_{n=0}^{\infty} n!\, x^n$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{n!} = n+1 \longrightarrow +\infty$$

So the radius of convergence $R = 0$, so the series is convergent only at it center.

2.) $\sum_{n=0}^{\infty} x^n$

$a_n = 1 \quad \forall n$, $\quad \frac{a_{n+1}}{a_n} = 1$

So the radius of convergence is $R = 1$.

3.) $\sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (= e^x)$

$$\frac{a_{n+1}}{a_n} = \frac{n!}{(n+1)!} = \frac{1}{n+1} \to 0$$

So the radius of convergence is $R = +\infty$.